# Analytical Chemistry 1 (125101)

Guangdong Technion
Israel Institute of Technology
广东以色列理工学院

# Analytical Chemistry1 (125101) (Credit 1.5 and 13 lectures)


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- Lecture, Tuesday by Shisong TANG
- Office time, tuesday, 10:00~11:30 @ E2-317 in SC
- Tutorials

  Dr. Marcos Tascon (T11 and T12)

  Dr. Satheeshkumar Elumalai (T13)

  Dr. Tareq Tarabeih (T21 and T22)

# Analytical Chemistry1 (125101) (credit 1.5 and 13 lectures)

- **Prerequisites:**

Fundamentals of Chemistry, Principle of Chemistry or General Chemistry (124114 or 124120 or 125001 or 125011)

- **Final Exams for Analytical Chemistry 1**

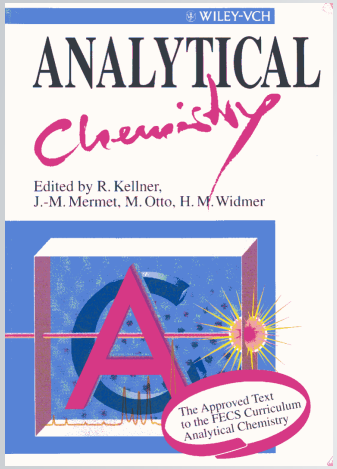
Term A: 14:00-17:00 June 24 (Tues.), 2025

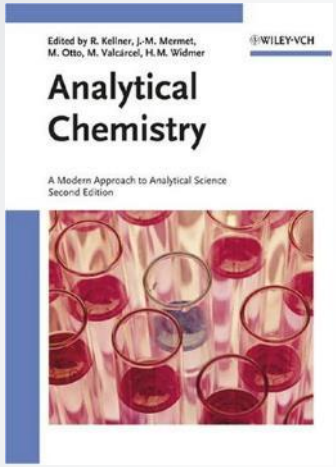
Term B: 14:00-17:00 July 9 (Wed.), 2025

- **Final course grade = Final exam grade ± *e***

"***e***" includes the bonus from good questions & deduction from lack of homework; e ≤ 10 points.

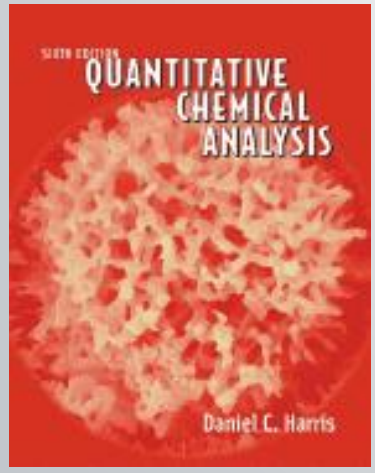
# Recommended textbooks

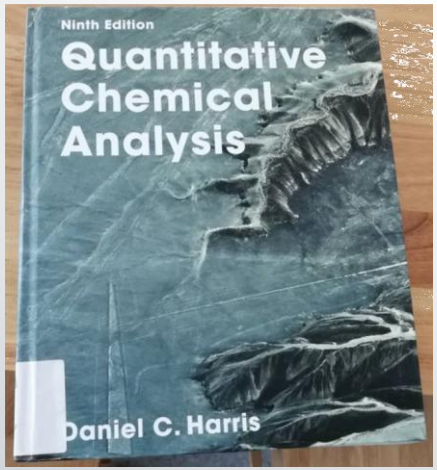



In the library of GTIIT

R. Kellner et al.
WILWY-VCH




Daniel C. Harris

W.H. Freeman and Co.

# Syllabus

- - **Introduction** of Analytical Chemistry, qualitative analysis and quantitative analysis.
- - **Experimental errors** (systematic and random). Estimation of experimental errors, normal distribution, estimation of mean and error from a finite number of measurements. Standard deviation of measurements and standard deviation of the mean.
- - **Propagation of errors:** The development of the general formula.
- - **Linear regression:** Development of the equations and discussing the assumptions, calibration methods, external standard, standard addition, internal standard.
- - **pH of acids and bases:** Exact mathematical description and approximations; Buffer solution, general Bronstead equation, exact mathematical description and application of chemical approximations.
- - **Titrations:** Mathematical description titration curve, precipitation, acid-bases, precipitation and redox titrations. Calculation and discussion of the titration accuracy, the methods to find end point by first derivative and Gran plot, Potentiometry.
- - **Electroanalysis:** Nernst equation, Voltammetry, Coulometry and Electrogravimetry.
- - **Gravimetric analysis:** Precipitation process and factors affecting the precipitation nature and its purity.

# 1. Introduction of

# Analytical Chemistry

# (Anal. Chem. or A.C. in abbreviation)

# 1. Introduction of Analytical Chemistry (A. C.) ?

1.1 What is A.C.?

1.2 What does product by A. C.?

1.3 What's the trends for A. C.?

1.4 Where does A. C. need?

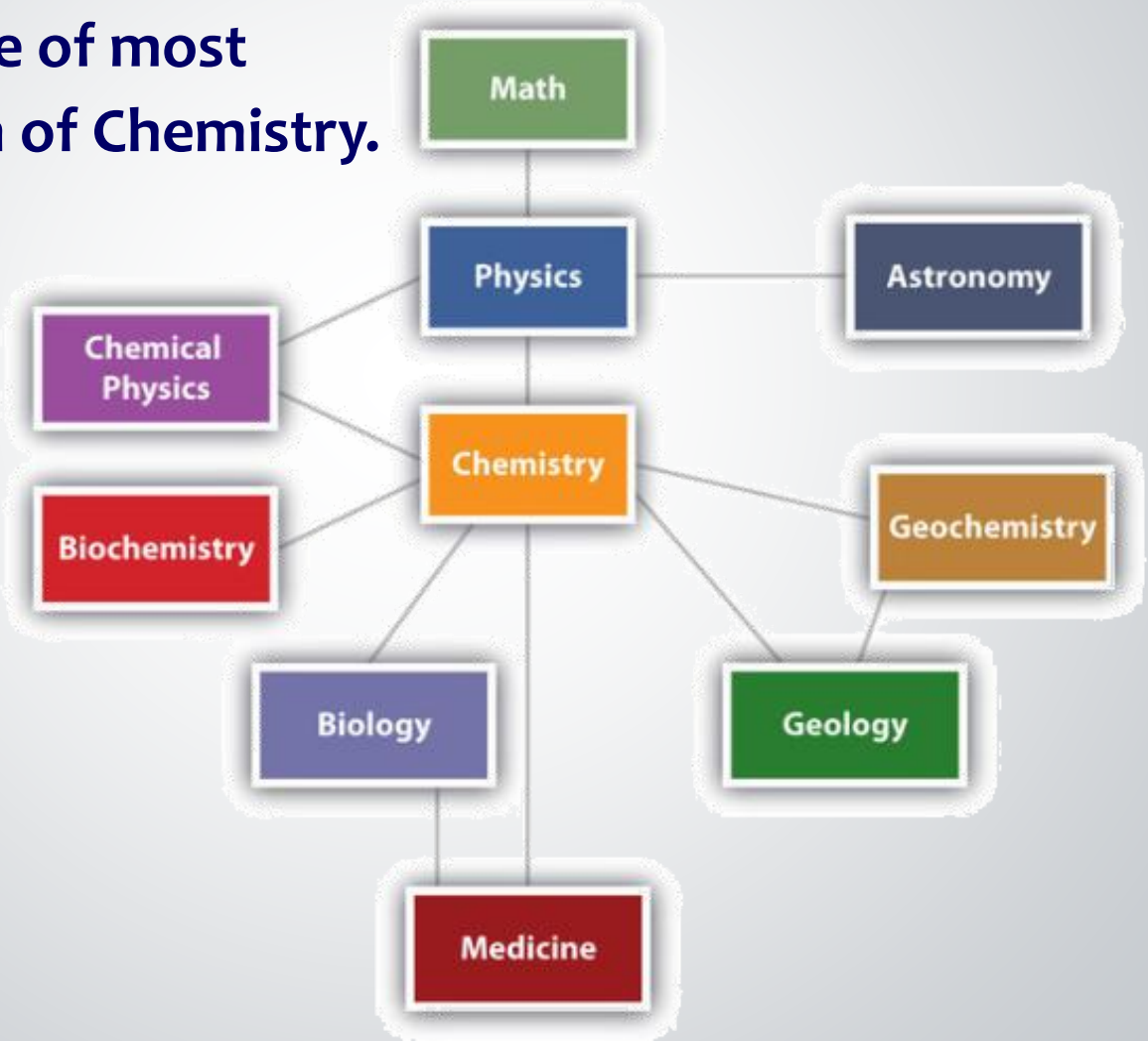
1.5 What is A. C. to do?

1.6 What's the general steps in A. C.?

1.7 What's the division of A. C.?

# 1.1 What is Anal. Chem.?

**Anal. Chem. is one of most important branch of Chemistry.**



# 1.1 What is Anal. Chem.?

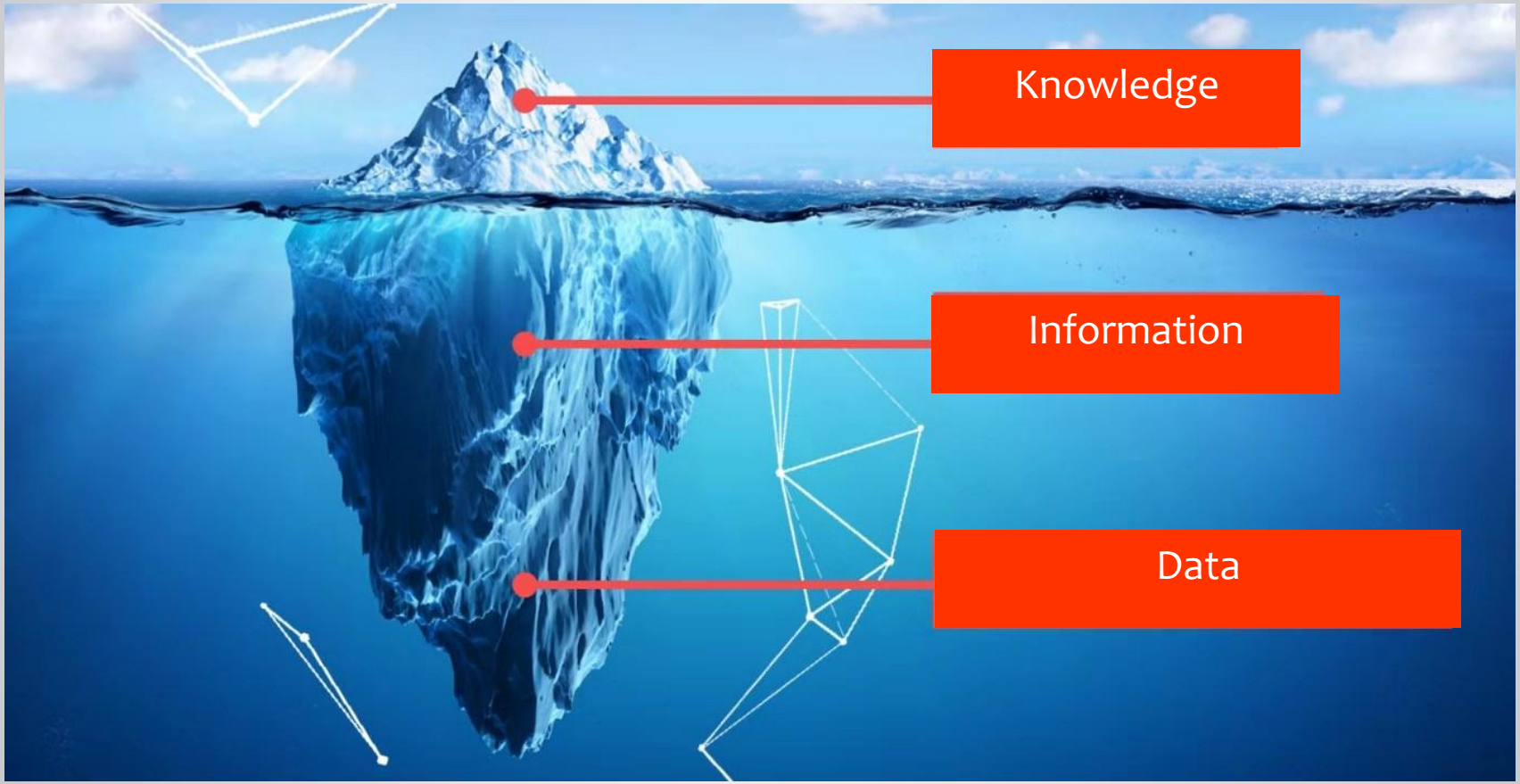
- Analytical Chemistry is a scientific discipline that develops and applies methods, instruments and strategies to obtain information on the composition and nature of matter in space and time.

— from FECS

FECS---Federation of European Chemical Societies

- **A-Chem. is an Information Science.**

# 1.2 What does product by Anal. Chem.?



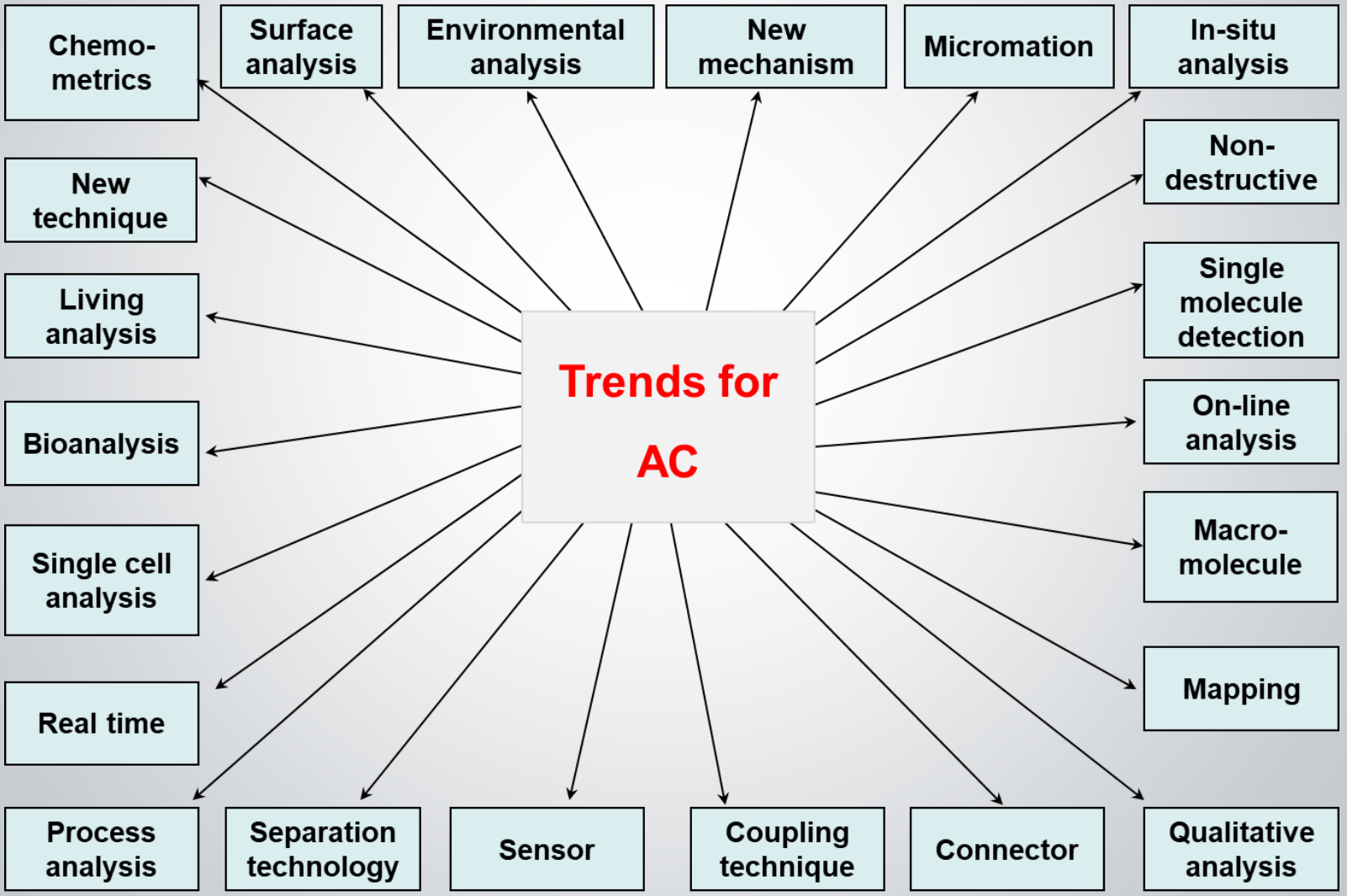

"Data-to-information" iceberg from C. Burgess, Valid

# 1.2 What does product by Anal. Chem.?

## Feature on Data, information and Knowledge

| | DATA | INFORMATION | KNOWLEDGE |
|---|---|---|---|
| **Definition** | Raw facts, figures and records contained in a system. | Data placed into a form that is accessible, timely and accurate. | Information in context to make it insightful and relevant for human action. |
| **Reason** | Processing | Storing / Accessing. | Insight, innovation, improvement. |

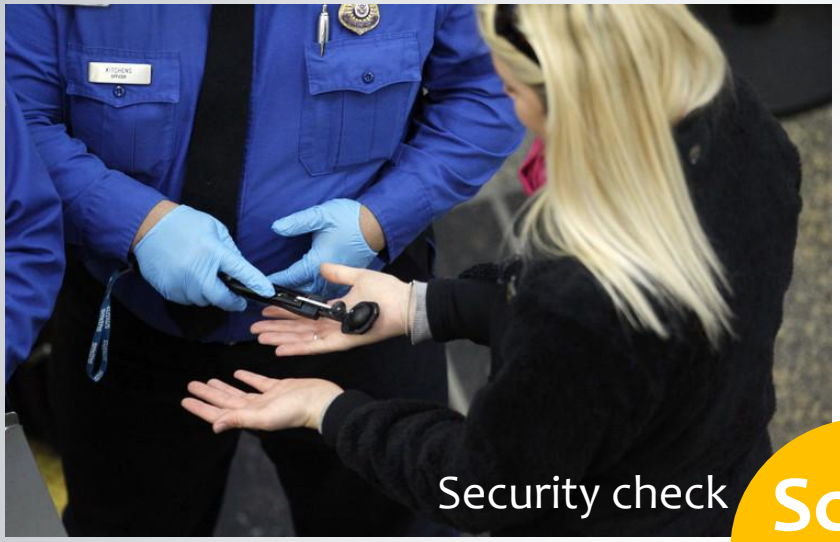
# 1.3 Trends for Anal. Chem.

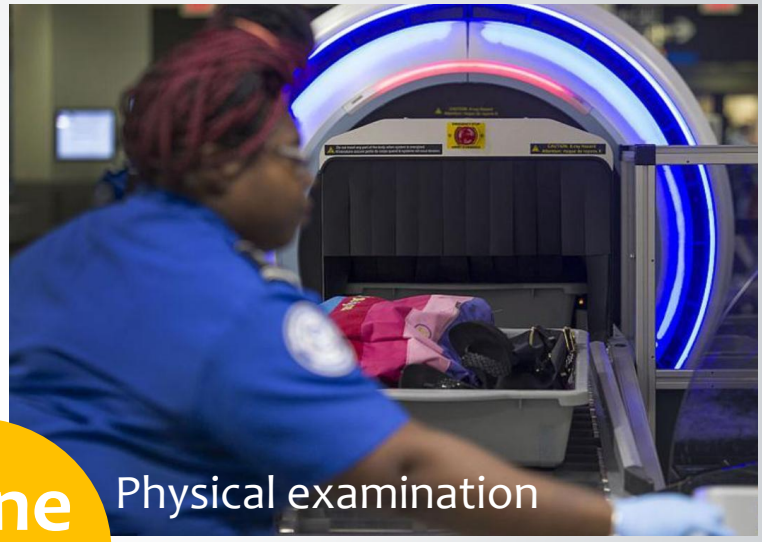


# 1.4 Where does Anal. Chem. need ?

| | |
|---|---|
| Chemistry | Biology |
| Food engineering | Pharmacology |
| Electronic industry | Quality control |
| Ecology | Archeology |
| Civil engineering | … |

# 1.4 Where does Anal. Chem. need ?




Scene demo

# 1.5 What is Anal. Chem. to do ?

- A-Chem. is the science to find the composition of matters.

  (1) Qualitative Analysis, what is it?

  (2) Quantitative Analysis, how much is it?

- ✓ Does this mineral contain enough copper to be an economical source?

- ✓ Home pregnancy test.

- ✓ How much caffeine is in the Chocolate bar?
- ✓ Is there sulfur element in human's hair?

# 1.5 What do Anal. Chem. do ?

- A-Chem. is the science of finding the composition of matter.

(1) Qualitative Analysis, what is it?

(2) Quantitative Analysis, how much is it?

(3) Structure Analysis

(4) Property Characterization

(5) Spatial Analysis --- mapping technology

Distribution of analyte through sample

(6) Time-dependent Analysis

Change of analyte/property over time

# General terms in "A-Chem. Talk"

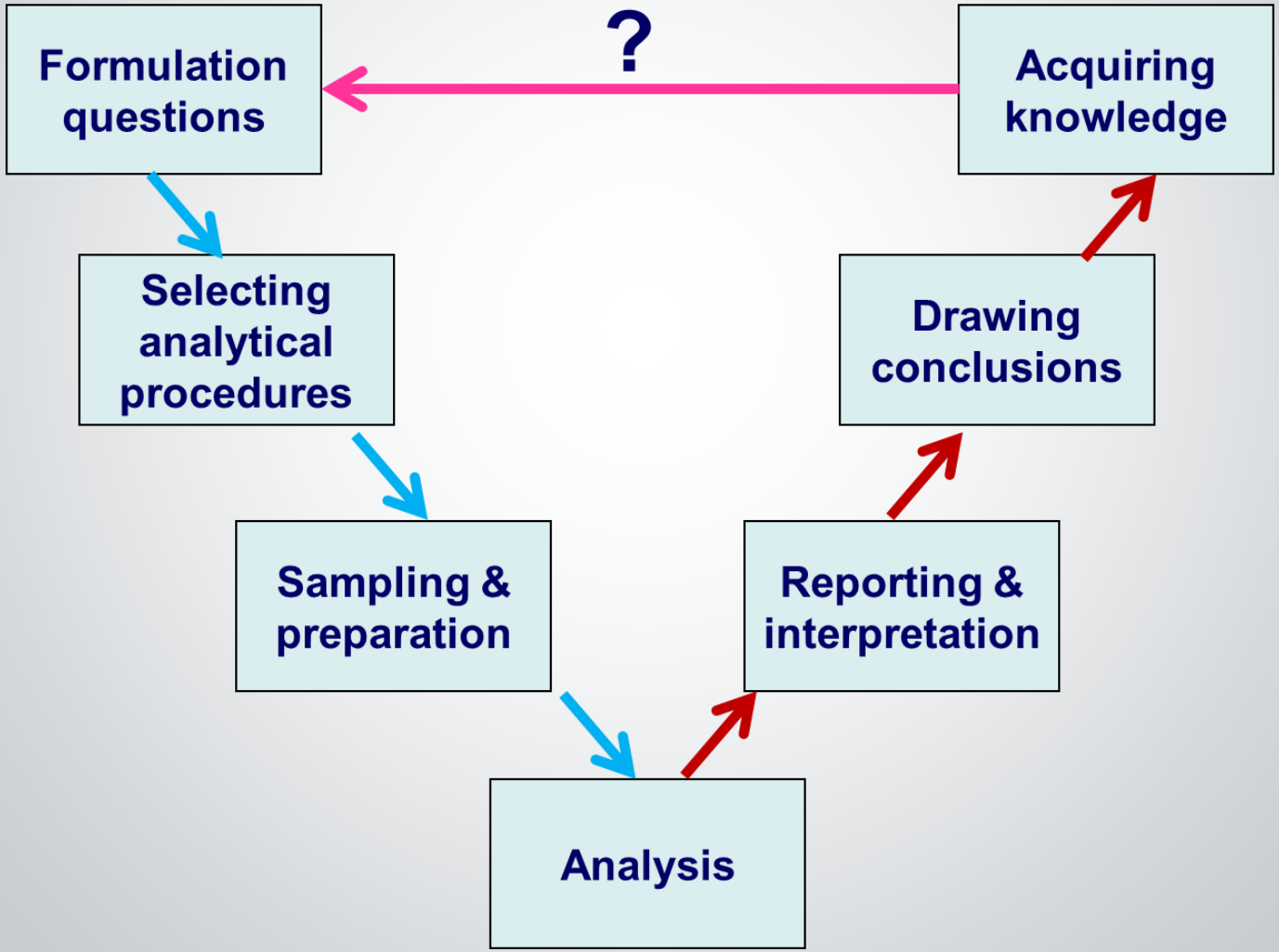
## Sample related terms

- Sample – representative of the tested material
- Matrix – all substance in the sample except the target
- Analyte – specific substance (target) we measure
- Major: > 1%
- minor: 0.01 % to 1%
- trace: below 0.01%

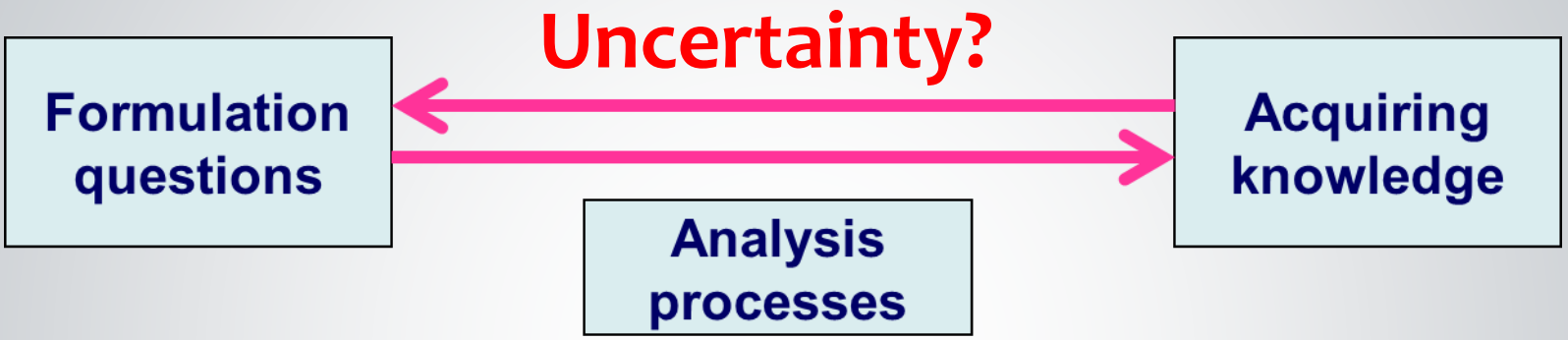
## Method related terms

- Analytical technique – an approach to the assay which involves
- Determining what question is being asked
- Using an appropriate sample
- Sample preparation
- Gathering measurements
- Reporting results

# 1.6 General steps in Anal. Chem.



# 1.6 General steps in Anal. Chem.




- Sampling, representative?
- Sample preparation, interference?
- Analytical method, reliable?

**The uncertainty of a measurement is as important as measurement itself in Analytical Science.**

distillation (boiling)

filtering

# Matter

separate *physically*

*cannot* separate *physically*

*Physical* changes

## Mixture

differences or unevenly mixed

uniform or evenly mixed

### *Heterogeneous* Mixture

dirt, blood, milk
(uneven, differences)

### *Homogeneous* Mixture

lemonade, gasoline, steel
(solutions)

## *Pure* Substance

separate *chemically*

*cannot* separate

*Chemical* changes

### Compounds

salt, baking soda, water, sugar

$NaCl$ $NaHCO_3$
$H_2O$ $C_{12}H_{22}O_{11}$

### Elements

oxygen, iron, hydrogen, gold

$O_2$ $Fe$
$H_2$ $Au$

# Types of Mixtures

- Heterogeneous mixtures
  - Can see the different parts of the mixture
  - Liquids are unstable and will settle out
  - EX. Pizza, Oil & water, etc.

- Homogeneous mixtures
  - Can NOT see the different parts of the mixture
  - Solutions
  - Liquids are stable and typically won't settle
  - EX. Alloys (metal mixtures), Salt water, etc.

- **Sample preparation**

- ✓ Dissolving
- ✓ Extraction
- ✓ Concentration
- ✓ Masking, interference not detected
- ✓ …

- **Analytical quality management(AQM)**
  - Construct different samples for analysis
  - Repeat measurements
  - Use several analytical methods
  - Measuring by different people
  - …

# 1.7 Division of Anal. Chem.

## Classical versus Instrumental

- Titration
- Gravimetry
- Precipitation
- Color reaction

- Electroanalysis
- Spectroscopy
- Separations
- Others

# Framework of this course

❖ Introduction (L1) No tutorial

❖ Section I Chemical Measurements

1.1 Basic knowledge (L2)

1.2 Error and estimation (L3&L4)

1.3 Linear regression (L5)

❖ Section II Chemical Equilibrium

2.1 pH and buffer solutions (L6)

2.2 Titrations (L7&L8)

2.3 Potentiometry in titrations (L9)

2.4 Gran plot in titrations (L10)

2.5 Electroanalysis (L11)

2.6 gravimetric analysis (L12)

❖ Summary (L13) No tutorial